THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A 20 cm object is placed 40 cm from a lamp and its shadow falls on a screen at 100 cm from the lamp. The shadow height is: 1

(a) 40 cm **(b)** 50 cm **(c)** 100 cm **(d)** 20 cm

2. In a dark room with a red light, a blue paper appears: 1

(a) white **(b)** red **(c)** black **(d)** blue

3. A girl stands 2 m from a mirror and a wall is 5 m behind her. The image appears: 1

(a) 5 m behind the mirror **(b)** 7 m behind

(c) 2 m behind the mirror **(d)** on the wall

4. A stick 1 m high casts a shadow of 0.8 m. A tower casts a shadow of 40 m at the same time. The height of the tower is: 1

(a) 50 m **(b)** 80 m **(c)** 32 m **(d)** 40 m

5. A white shirt in the red light appears: 1

(a) blue **(b)** red **(c)** white **(d)** black

6. A ray reflected by a plane mirror makes an angle of 100° with the incident ray. The angle of incidence is: 1

(a) 80° **(b)** 100° **(c)** 25° **(d)** 50°

7. A boy 1.5 m tall casts a shadow 2 m long. A tree casts a 12 m shadow. The tree height is: 1

(a) 12 m **(b)** 8 m **(c)** 16 m **(d)** 9 m

8. Why can we read a book by the reflected light from a lamp? 1

(a) paper reflects light diffusely **(b)** paper emits light

(c) paper is a mirror **(d)** paper is transparent

9. If an object is placed in front of two mirrors inclined at 90°, the number of images is: 1

(a) 4 **(b)** 1 **(c)** 3 **(d)** 2

10. A small lamp is 1 m above the ground. A ball of diameter 10 cm is placed 0.5 m below the lamp. The shadow diameter on the ground is: 1

(a) 20 cm **(b)** 5 cm **(c)** 40 cm **(d)** 10 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain how a pinhole camera could be used to observe a solar eclipse safely. 2

12. Give a reason why a periscope's mirrors are plane. 2

13. A student at 3 m from a plane mirror moves to 5 m. By how much does the distance between her and her image change? 2

14. A girl wishes to see her full image in a plane mirror. What is the least mirror height needed? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A 6 cm candle is placed 30 cm from a pinhole of a camera whose length is 15 cm. Find the image height. 3

16. Why do you see your face upside down in the inner side of a polished spoon? 3

17. How would you demonstrate lateral inversion with a piece of paper? 3

OR

A stick 1.2 m casts a shadow 0.9 m. At the same time a building casts a shadow of 30 m. Find the height. 3

18. Explain how a teacher can show that the image in a plane mirror is virtual. 3

OR

Explain why a shadow is not formed by a perfectly transparent glass but a clear glass may produce a faint shadow. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A boy 1.6 m tall stands 3 m from a vertical plane mirror. Find the minimum height of mirror required to see his full image and where its top edge must be, relative to the boy's eyes (eyes 10 cm below top of head). 5

OR

Explain how reflection and shadows are used in photography (e.g. umbrella reflectors) and traffic safety (reflectors). 5

20. Explain, with the help of the laws of reflection and ray diagrams (in words), how a periscope gives an erect image and how the length of the tube may be changed. 5

OR

A circular disc of diameter 5 cm is placed 30 cm from a point source and a screen is 90 cm from the source. Find the diameter of the shadow, and the area ratio of shadow to disc. 5

END OF QUESTION PAPER — Total: 40 Marks

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